Wireless Communication Exam 1

Note**

1. Each question carries 10 marks
2. Follow the format for submission

Student Name: jkl

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Student ID: 321

1. Define modulation and explain its purpose in wireless communication.

Answer: Modulation is the process of increasing the power of a signal to ensure it can travel longer distances without attenuation. Its primary purpose in wireless communication is to amplify signals for better reception.

Score: 0/0

Justification: N/A

Feedback: No feedback provided

2. Calculate the path loss at a distance of 500 meters for a signal frequency of 2 GHz in free space.

Answer: The path loss at 500 meters for a 2 GHz signal in free space is approximately 50 dB .

Score: 1.5/10

Justification: The student's answer lacks explanation, is incorrect, and demonstrates a lack of understanding of path loss and its calculation. The calculated path loss is incorrect, and the answer is not well-organized.

Feedback: The student should provide a clear explanation of path loss's impact on signal strength and reliability, double-check calculations, and organize their answer logically. They should review the concept of path loss, practice applying the formula, and ensure correct application of values.

Summary:

Total Score: 1.5

Percentage: 7.50%

Grade: F

Overall Feedback: To improve, the student should focus on providing a clear explanation of path loss's impact on signal strength and reliability, double-checking calculations, and organizing their answer logically. They should also review the concept of path loss, practice applying the formula, and ensure correct application of values.